MACHINE LEARNING FOUNDATIONS · FINAL ASSESSMENT

Final Project: A Supervised Model for Pinecrest

The end-of-course assessment of Machine Learning Foundations. Each learner chooses one of three supervised problems on the Pinecrest data (churn, weekly demand or fraud), builds one model end to end with the course's leak-safe split, a baseline, a fair comparison, honest tuning and one look at the test set, and reports the decision in one page. Score it with the seven-row rubric below. This page also holds the instructor answer notes with reference numbers for all three problems: it is for instructors only.

Covers **Modules 1–6** | Chosen **Session 10**, studio **Session 11** | Submit within **1 week** | Successful: **3+** on every row | Instructor only

THE BRIEF (AS THE INSTRUCTOR SEES IT)

Pinecrest's analytics lead, in the words used in Session 10: "Pick one of three problems we actually have. Build one supervised model end to end, the way this course taught: frame the decision, split the data the way the model will be used, beat a do-nothing baseline, compare candidates fairly, tune without fooling yourself, open the test set once, and tell us in one page what we should do and how sure you are. An honest 'the simple model is enough' is a perfectly good answer."

PROBLEM	DATA AND THE LEAK-SAFE SPLIT (FIXED: EVERYONE USES IT)	DECISION AND MEASURES	DO-NOTHING BAR
A · Churn: who gets a retention offer?	`customers_churn.csv`; `train_test_split(test_size=0.25, stratify=churned, random_state=42)`: 4,500 / 1,500 (248 test churners). Never `cancellation_reason`.	Offer or not. $15 per offer, $60 per churner reached (Lesson 1.4). Choose models by average precision (or ROC AUC); decide by profit at a threshold chosen on training rows.	Contact everyone -$7,620; nobody $0; random-order AP 0.165

PROBLEM	DATA AND THE LEAK-SAFE SPLIT (FIXED: EVERYONE USES IT)	DECISION AND MEASURES	DO-NOTHING BAR
B · Weekly demand: how many units to stock?	`weekly_demand.csv`; train `year == 2024` (5,088 rows, 53 week starts), test 2025 (4,896). CV folds are blocks of weeks (`KFold(5)` without shuffling).	Units per product, store and week, planned at the start of 2025 (no 2025 sales figure may be a feature). MAE in units, overall and by category.	Per-product 2024 mean MAE 13.42; overall mean 17.13
C · Fraud: which payments to review?	`transactions.csv`; train January to May (41,738 payments, 374 frauds), test June (8,262, 64 frauds). CV inside January to May.	Review or not. $10 per review, $375 per missed fraud (Lesson 2.5). Choose models by average precision; decide by June cost at a threshold chosen on January to May.	Review nothing $24,000; everything $82,620; "always legitimate" accuracy 0.9923

A learner may propose another supervised problem on the course data (for example review sentiment from `reviews.csv`) if it names a decision, has a leak-safe split matching how the model would be used, and is not a lesson project re-run. Approve it in writing before Session 11.

- **Deliverables:** `final_project.ipynb` (Restart and Run All reproduces every number from the project folder), `report.md` (one page: Question, Answer, Evidence, Caveats, Next questions, Method), `README.md` (how to rerun, what help was used), `results.csv` (the leaderboard: candidate, mean, std of the deciding metric), `best_params.json` (if a search was run). Optional: one figure with a finding-first title.
- **Timeline:** chosen and planned in Session 10 (20-minute segment); milestone 1 (split, bar to beat, first CV score; `labs/module-11/l6_2-hyperparameter-tuning/final_project_starter.py`) before Session 11; milestones 2 and 3 (fair leaderboard, tuning) in the Session 11 studio; milestones 4 and 5 (test once, decision, report) at home; submit within one week of Session 11. Expected effort: about 6 to 8 hours in all.
- **Assessed:** the submitted folder against the seven rubric rows. The Session 11 showcase is practice and is not scored; note one strength per learner from it for the written feedback.
- **There is no answer key for learners.** The notes below are for coaching and grading. An honest result that the simple or default model is enough scores exactly as well as a gain: for demand, tuning does not beat the defaults; for fraud, the good models are within one fraud of each other in June.

THE RUBRIC

CRITERION	1 · BEGINNING	2 · DEVELOPING	3 · PROFICIENT	4 · EXEMPLARY
Framing and metric weight 10%	A target and a model, no decision. The metric is accuracy, or is never stated.	A decision is named, but the metric is chosen after seeing results, or does not match the decision (accuracy for a rare class).	The decision, who acts on it, the moment of prediction, the deciding metric and the business measure (profit, MAE in units, cost) are written down before the analysis and the work stays on them.	Also states in advance what result would change the recommendation, and revises the plan openly when the data require it.
Leak-safe data and split weight 20%	A random split for demand or fraud, or a leaking column in X (`cancellation_reason`, a 2025 sales figure, anything written after the outcome).	The course's split is used, but preprocessing is fitted outside a pipeline, or a feature's availability on prediction day is never checked.	The course's split exactly (churn stratified 75/25 random_state 42; demand 2024 / 2025; fraud January to May / June); every preprocessing step inside a pipeline; each feature justified as known at prediction time; cross-validation folds that respect time where it matters.	Also shows what a tempting leak or a shuffled split would have reported, and why that number is not the result.
Baselines weight 10%	No baseline, or only a training-set score to compare with.	A baseline is computed but is not the right bar (a model's training score; for demand, zero).	The do-nothing bar in the deciding metric AND in business terms: random-order AP and offering to everyone (churn), the per-product mean MAE (demand), reviewing nothing (fraud). Every later number is compared with it.	Also includes a second, smarter simple rule (per product and store, same week last year, a hand-made rule) and explains what the model adds over it.

CRITERION	1 · BEGINNING	2 · DEVELOPING	3 · PROFICIENT	4 · EXEMPLARY
Fair model comparison weight 15%	Models compared on one test-set score each.	Cross-validation, but different folds or different metrics per model, or a winner crowned by the third decimal.	Baseline + at least two real candidates as pipelines on ONE splitter, the deciding metric, mean and std side by side, a paired per-fold check of the top two with a stated conclusion (real / within noise).	Also a learning or validation curve that justifies the next step, or a feature idea tested on the same folds.
Tuning and honest estimates weight 15%	Settings changed by hand while watching the test score.	A search on the training rows, but a linear grid for log-scale knobs, no written space, or best_score_ reported as the model's performance.	A written space (log scales for rates and strengths), a stated budget, a search on the training rows only; best_score_ used only to choose; an honest estimate (nested CV or the one-time test score) reported; an honest negative if tuning does not help.	Also reports how many settings were within noise of the winner, and what they have in common.

CRITERION	1 · BEGINNING	2 · DEVELOPING	3 · PROFICIENT	4 · EXEMPLARY
Test once and the decision weight 15%	The test set opened repeatedly, or used to choose the model or the threshold.	Opened once, but the result is reported only as a model metric, with no decision.	The chosen model fitted on all training rows and scored once on the test set; the decision in business terms (churn: offers, churners reached, profit at a threshold chosen on training rows; demand: MAE in units overall and by category; fraud: queue size, frauds caught and cost at a threshold chosen on January to May), compared with the do-nothing policy.	Also quantifies the uncertainty of the test result (a bootstrap, or the CV spread) and says which differences are within it.
Report and reproducibility weight 15%	No report, or the notebook fails on rerun.	A report that retells the steps, numbers without their comparison, or a notebook that depends on cell order or unseeded randomness.	One page (Question, Answer, Evidence, Caveats, Next questions, Method) that leads with the answer and quotes the honest numbers; Restart and Run All works from the project folder with relative paths and seeds; every number in the report appears in the output.	Also a README, saved settings (`best_params.json`) and results files, and caveats specific to the data (drift, stockouts, unseen growth) a non-specialist could act on.

WEIGHTING AND THE RESULT

- Score each row 1 to 4 (whole numbers). The weighted score is the sum of score × weight, out of 4.00. Weights: Framing and metric 10%, Leak-safe data and split 20%, Baselines 10%, Fair model comparison 15%, Tuning and honest estimates 15%, Test once and the decision 15%, Report and reproducibility 15%.
- **Successful final project:** 3 (Proficient) or higher on every row. Report the weighted score alongside it.
- **Successful with a revision:** one row at 2 and a weighted score of at least 2.75; the learner fixes that row within one week and you re-score that row only.
- **Not yet:** any row at 1, two or more rows at 2, or a notebook that does not run. Return written feedback (one thing done well per row and the single change that would most improve the work) and allow one resubmission within two weeks.
- **Leakage overrides polish:** a model trained with a leaking column or a split that mixes the test period into training scores at most 2 on "Leak-safe data and split" AND on "Test once and the decision", however good the numbers look.
- **Reproducibility first:** rerun the notebook before reading anything else. If it fails, ask for a fix within 48 hours before scoring the other rows.

SCORING GUIDANCE BY ROW

ROW	WHAT TO LOOK FOR
Framing and metric	The first Markdown cell and the report's Question line: a decision, a moment of prediction, a deciding metric and a business measure, written before results. Accuracy as the deciding metric for churn or fraud caps this row at 2.
Leak-safe data and split	Split sizes must match the brief exactly (churn 4,500 / 1,500 with first test id C102112; demand 5,088 / 4,896; fraud 41,738 / 8,262). Search X for `cancellation_reason`, `transaction_id`, `timestamp`, `year`, lagged 2025 sales. Preprocessing inside pipelines; demand CV without shuffling.
Baselines	The do-nothing number in both the metric and business terms, computed in the notebook (not typed). A second simple rule earns 4.
Fair model comparison	One splitter object passed to every candidate; mean and std; a paired comparison of the top two with a sentence ("real" or "within noise"). `results.csv` matches the notebook.
Tuning and honest estimates	A space with log scales, a budget, the search on training rows. Which number is reported as performance: best_score_ alone is Developing. A clearly reported "tuning did not help; default kept" is Proficient.

ROW	WHAT TO LOOK FOR
Test once and the decision	One cell opens the test set, after the choice. The threshold (churn, fraud) was chosen on training rows (TunedThresholdClassifierCV or out-of-fold profit curve). The decision is compared with the do-nothing policy in dollars or units.
Report and reproducibility	Read `report.md` first, as the analytics lead would: answer first, honest numbers, specific caveats, about one page. Then rerun and find three report numbers in the output.

SCORING SHEET

Learner: _____ · Problem: A / B / C / own · Date received: _____ · On time: yes / no

ROW	WEIGHT	SCORE 1–4	SCORE × WEIGHT	EVIDENCE (CELL OR LINE IN REPORT.MD)
Framing and metric	10%			
Leak-safe data and split	20%			
Baselines	10%			
Fair model comparison	15%			
Tuning and honest estimates	15%			
Test once and the decision	15%			
Report and reproducibility	15%			
Weighted total	100%			Result: successful / successful with revision / not yet

SUBMISSION AND ACADEMIC HONESTY

- **Deadline:** one week after Session 11, as a zipped project folder without the course `data/` folder, or a link to a repository.
- **AI assistants and code from the web** are allowed on the course's terms: you are responsible for every line and every claim. The README says what help was used and for what. You may pick any two cells and ask the learner to explain them in five minutes; a cell they cannot explain is revised before the row it affects is scored.

- **Discussion is encouraged; shared code is not.** Learners on the same problem should compare split sizes and baselines (they must match); notebooks and wording are their own.
- **Data:** the course data are fictional (Pinecrest Outfitters) and belong to the course. No real or work data, no secrets in any cell.
- Do not share these notes with learners; feedback refers to the rubric and to the learner's own work.

LATE WORK AND ALTERNATIVES

- **Extension:** one extra week on request before the deadline, no reason needed.
- **Late without an extension:** still scored on the rubric, with the date recorded; no points deducted.
- **Missed Session 11:** a 15-minute check-in with you within the week replaces the studio coaching; the showcase is replaced by a written five-line card (decision, bar, best CV ± spread, one trap, what is left).
- **Narrower scope agreed in advance:** two candidates instead of three and no search; the tuning row is then scored on the honest-estimate part only.

RECORDING CHECKPOINT 1 ALONGSIDE THE FINAL PROJECT

Checkpoint 1 (Session 6, Modules 1 and 2, its own five-row rubric in the Session 6 answer key) is formative and is not weighted into the final project. Course completion = Checkpoint 1 met (or redone) and a successful final project. A Not yet on Checkpoint 1 predicts trouble in the "Leak-safe data and split" and "Fair model comparison" rows: visit those learners first in the Session 11 studio.

LEARNER	CHECKPOINT 1 (S6)	PROBLEM	ROWS (F, D, B, C, T, E, R)	WEIGHTED	RESULT

ANSWER NOTES · PROBLEM A: CHURN

A SOUND ANALYSIS: STEPS

- The 11 numeric and 5 categorical features of Module 2 (`cancellation_reason` out), the canonical split, preprocessing inside pipelines (median / most-frequent impute, scale, one-hot; or native categories for HistGradientBoosting).
- Bar to beat: random-order AP 0.165 (= the test churn rate), offering to everyone, and offering to nobody ($0).
- Candidates on one `StratifiedKFold(5, shuffle=True, random_state=42)` (or the 6.1 repeated splitter), deciding metric average precision (the care team works from the top of a ranked list); a paired check of

the top two.

- A random search over boosting (the 6.2 space: loguniform rates and L2, randint leaf counts and leaf sizes, 10 to 15 settings) on the training rows.
- The threshold for the $15 / $60 economics chosen on the training rows (TunedThresholdClassifierCV with a profit scorer, or an out-of-fold profit curve, Lesson 1.4); the test set opened once for the chosen model and threshold.

KEY NUMBERS (FROM THE INSTRUCTOR'S OWN RUN ON THE COURSE DATA)

- 5-fold CV average precision on the training rows: logistic regression 0.528 ± 0.042 (AUC 0.832), random forest (200 trees, leaf 5) 0.619 ± 0.042, default HistGradientBoosting 0.608 ± 0.043 (AUC 0.861), baseline 0.165.
- Tuned boosting (15 random settings, AP): best_score_ 0.636 with learning_rate 0.0472, max_leaf_nodes 50, min_samples_leaf 199, l2 0.0292. Test, once: AP 0.614, ROC AUC 0.873 (default: 0.6 / 0.86).
- Churners in the top 150 test customers (the care team's quarterly calls): logistic 87, forest 97, default boosting 101, tuned boosting 100; a random list reaches about 25. Tuning changes the ranking quality a little and the calls not at all.
- Logistic regression with four shape features (lapsed > 150 days, tenure < 6 months, tenure > 72 months, tickets × no app): CV AP 0.643 (AUC 0.878), test AP 0.621 (AUC 0.874): as good as tuned boosting, from features alone.
- Threshold chosen on the training rows for the tuned booster: 0.222 (break-even 0.25). Test, once: 395 offers, 186 reach a churner, profit $5,235; the same model at 0.5 earns $3,780, at a fixed 0.25 $5,400; offering to everyone -$7,620. Logistic regression with its own tuned threshold (0.233): $4,080 with 352 offers.

THE ANSWER THE DATA SUPPORT

Offer the retention deal to customers whose predicted churn probability is above about 0.22 to 0.25: on 1,500 held-out customers that is about 395 offers reaching about 186 real churners, an expected profit of about $5,235 against a loss of $7,620 for offering to everyone and $3,780 at the default 0.5. Most of the gain over the default comes from the threshold, not from tuning; a tuned booster and a feature-built logistic regression are equally good choices, and the simpler one is easier to explain. The dollar figures rest on the $15 and $60 estimates.

TRAPS (AND WHAT THEY REPORT)

- `cancellation_reason` in X: test ROC AUC 0.939, AP 0.866. Too good to be true: it is filled in only after a customer leaves.
- Accuracy as the deciding metric: doing nothing scores 0.835; a model at 0.85 sounds weak and is useful, or sounds strong and is useless.
- The threshold chosen on the test set: the best test threshold (0.27) shows $5,535, about $300 more than the honest $5,235; it is a best-of-many on the grading data, not a result.
- best_score_ (0.636) reported as the model's performance: the one-time test AP is 0.614.

- Several test looks: comparing every candidate on the test set and keeping the best. The leaderboard belongs to the training rows.

WHAT AN HONEST RESULT SOUNDS LIKE

- "Tuning raised cross-validated AP from 0.608 to 0.636 (best of 15 settings, so optimistic), and test AP from 0.600 to 0.614; in the care team's top 150 it made no difference (100 against 101 churners). The decision that matters is the threshold: chosen on the training customers at 0.22, it earns about $5,200 per 1,500 customers."
- Follow-up questions: "Which rows chose your threshold?" · "How many more churners does your model put in the top 150 than the simplest model?" · "How would the plan change if an offer cost $20?"
- Weak submissions: report accuracy; include `cancellation_reason`; quote best_score_ as performance; give a profit with no comparison to offering to everyone.

ANSWER NOTES · PROBLEM B: WEEKLY DEMAND

A SOUND ANALYSIS: STEPS

- Features known when the 2025 plan is made: store, product, category (category dtype), week of year, list price, discount, price, promo, holiday week, temperature (as if a seasonal forecast were available: a caveat to state). Not the year, not any 2025 sales.
- Train on 2024 (5,088 rows: 53 week starts, because 2024-12-30 belongs to 2024), test on 2025 (4,896).
- Baselines: the 2024 overall mean, each product's 2024 mean, per product and store, the same week last year.
- CV inside 2024 on blocks of weeks (`KFold(5)` without shuffling), MAE in units; a Poisson loss for counts (Lesson 5.3).
- A random search on the blocks; the default kept if it is not beaten; 2025 opened once; MAE overall and per category; the total forecast against the total sold.

KEY NUMBERS (FROM THE INSTRUCTOR'S OWN RUN ON THE COURSE DATA)

- 2025 MAE of the simple rules: overall mean 17.13, per-product mean 13.42, per product and store 12.95, same week last year 13.81.
- HistGradientBoostingRegressor, native categories: squared error 8.16, Poisson 8.05 (R^2 0.744): 40% below the per-product mean.
- Cross-validation inside 2024 for the Poisson model: shuffled KFold 7.6 ± 0.19 (flattering), blocks of weeks 8.49 ± 1.34, TimeSeriesSplit 9.77 ± 1.04 (early folds train on very little of the year).
- A 15-setting random search on the blocks: best CV MAE 8.57 (WORSE than the default's 8.49); its 2025 MAE 8.22 against 8.05. Keep the default.
- Per category (Poisson boosting, 2025, baseline in brackets): Snow 4.45 (baseline 13.33); Packs 5.5 (baseline 9.16); Climbing 5.71 (baseline 9.11); Sleeping 6.63 (baseline 10.69); Tents 6.82 (baseline 11.27); Apparel 8.08

(baseline 12.59); Accessories 8.86 (baseline 13.07); Camp Kitchen 11.61 (baseline 19.31); Footwear 12.85 (baseline 18.72).

- Level: mean forecast 24.6 units per row against 26.3 sold in 2025 (24.4 in 2024): the total forecast is 121,456 units against 128,970 sold, about 6% low, because trees cannot extrapolate growth they never saw.
- Zeros: 105 weeks sold 0 in 2025; 5 of them had a forecast above 12 units (likely stockouts, which a sales history records as zero demand).

THE ANSWER THE DATA SUPPORT

Default Poisson gradient boosting forecasts 2025 weekly units with an MAE of about 8.05 units against 13.42 for each product's 2024 average: the biggest gains are where the season matters (Snow 4.45 against 13.33). Tuning did not improve it. The forecast is about 6% low overall because 2025 grew; a buyer should scale it up or the model should be retrained as 2025 data arrive.

TRAPS (AND WHAT THEY REPORT)

- A shuffled 75/25 split over both years: MAE 7.36, better than the honest 8.05, because 2025 weeks leak into training.
- Shuffled CV folds inside 2024: 7.6, which then makes every tuning comparison look rosier than 2025 will.
- Last week's units as a feature: MAE 8.17 (no better here), and invalid for a plan made at the start of the year; valid only for a one-week-ahead forecaster, a different project.
- `year` as a feature: MAE 8.05 (no effect: trees cannot extrapolate a year they never saw), and it hides the growth problem rather than solving it.
- Reporting RMSE or R^2 only: the buyers think in units; MAE in units, by category, is the business measure.

WHAT AN HONEST RESULT SOUNDS LIKE

- "A search over 15 settings did not beat the default settings (CV MAE 8.57 against 8.49 on blocks of 2024 weeks), so we kept the default; on 2025 it is off by 8.05 units a week on average, 40% better than each product's 2024 average." That is a Proficient tuning row.
- Follow-up questions: "On 2 January 2025, which of your features would you not know?" · "Why are your shuffled-fold scores better than your 2025 score?" · "Is the forecast too high or too low in total, and why?"
- Weak submissions: a random split; R^2 only; a tuned model that is worse but reported because it was tuned; no per-category view.

ANSWER NOTES · PROBLEM C: FRAUD

A SOUND ANALYSIS: STEPS

- Nine numeric columns plus `category`; not `transaction_id` or `timestamp` (order, not behavior); train January to May, test June.

- Bar to beat in June: review nothing, review everything; "always legitimate" accuracy shows why accuracy is useless here.
- Candidates on stratified folds inside January to May, deciding metric average precision: logistic regression (scaled, one-hot), default and regularized HistGradientBoosting (Lesson 5.3 Exercise 4 settings), optionally a forest.
- A random search over boosting with early stopping; then a review threshold for $10 / $375 chosen on January to May (TunedThresholdClassifierCV with a cost scorer; the formula gives 10 / 375 = 0.027 if probabilities are calibrated).
- June opened once: queue size (per day), frauds caught, cost, against reviewing nothing.

KEY NUMBERS (FROM THE INSTRUCTOR'S OWN RUN ON THE COURSE DATA)

- 5-fold CV average precision on January to May: logistic 0.506 ± 0.054 (AUC 0.855), default boosting 0.443 ± 0.033 (overfits 374 frauds), regularized boosting 0.528 ± 0.048; baseline about 0.009.
- Tuned boosting (15 settings, early stopping): best_score_ 0.529 (learning_rate 0.1011, max_leaf_nodes 12, min_samples_leaf 176, l2 2.7527). June, once: AP 0.565, 39 of 64 frauds in the 100 highest scores (logistic 0.539 / 36; regularized 0.556 / 36; default 0.477 / 32).
- Threshold chosen on January to May, June cost: logistic 0.051 → 166 reviews, 41 caught, $10,285; regularized boosting 0.03 → 172, 41, $10,345; tuned boosting 0.04 → 165, 42, $9,900. Review nothing $24,000; the default 0.5 threshold costs $15,970 (logistic).
- Night features (hour < 6, night × new device): logistic CV AP 0.532, June cost at a tuned threshold $10,120; boosting 0.53: small changes, within noise.
- About 5.5 reviews a day in June at any of these thresholds (8,262 payments over 30 days).

THE ANSWER THE DATA SUPPORT

A review queue of about 165 to 190 June payments (about 6 a day), chosen by a model trained on January to May with a threshold tuned for the $10 / $375 costs, catches about 41 or 42 of 64 frauds and costs about $9,900 to $10,300 against $24,000 for reviewing nothing. Tuned boosting, regularized boosting and logistic regression are within a few hundred dollars of each other: about one fraud ($375), so June cannot rank them; choose the simplest one you can explain and monitor, and expect the fraud rate to keep drifting (1.00% in January, 0.77% in June).

TRAPS (AND WHAT THEY REPORT)

- Accuracy: reviewing nothing is 0.9923 accurate in June and catches no fraud.
- A threshold tuned on June: the best June threshold for tuned boosting (0.03) shows $9,395, about $500 better than the honest $9,900: chosen by the data that grades it.
- A stratified random split of all six months: AP 0.506 on 131 test frauds. The number is not dramatically inflated here, but the model is tested on months it trained around and the drift is hidden; the split does not match how the model is used.

- Default boosting: CV AP 0.443: with 374 positives the defaults overfit; gentle settings or a search fix it.
- Claiming a winner by a few hundred dollars: one fraud is $375 and June has 64.
- `transaction_id` or the timestamp as a feature: ordering, not behavior (here it changes little: CV AP 0.524), and it breaks as soon as the model scores July.

WHAT AN HONEST RESULT SOUNDS LIKE

- "Trained on January to May, with the review threshold chosen on those months, the queue would have held 165 June payments (5.5 a day), caught 42 of 64 frauds and cost $9,900 against $24,000 for reviewing nothing. Logistic regression did almost as well ($10,285); the difference is about one fraud, so we cannot say which is better from June alone."
- Follow-up questions: "Which months chose your threshold?" · "How many reviews a day does your queue need?" · "What would you watch in July to know the model still works?"
- Weak submissions: accuracy; a random split; a June-tuned threshold; a winner declared on a $300 difference.

HOW THESE NOTES WERE MADE

Every number above comes from the instructor's own run on the shipped course data (scikit-learn 1.9.1, pandas 3.0, 4 CPU threads, `random_state=42` everywhere), using the scripts `kitA_measure_churn.py`, `kitA_measure_demand.py` and `kitA_measure_fraud.py` in the kit's generator folder. Learners will reach slightly different numbers with other reasonable choices (other candidates, another search budget, another threshold method): judge the split, the comparison and the honesty of the report, not agreement to the third decimal. The split sizes and the do-nothing numbers must match exactly.

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